

Postpartum Hemorrhage 06: Massive Transfusion, Coagulopathy, and DIC

Companion reading handout for simultaneous listening.

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Massive transfusion, coagulopathy, and DIC.

In the last episode, we discussed escalation.

Bakri presses from inside.

B Lynch compresses from outside.

Uterine artery ligation lowers inflow.

Embolization closes selected vessels.

Hysterectomy removes the bleeding organ.

Now we focus on the blood

Not only the blood that is leaving the patient, but the blood products, clotting factors, temperature, electrolytes, and physiology that determine whether source control can succeed.

The central idea is this:

Coagulopathy treatment is not after source control.

It happens alongside source control.

If the surgeon is trying to stop bleeding while the patient is cold, acidotic, calcium-depleted, and fibrinogen-depleted, the field may keep dissolving in front of them.

Source control and resuscitation are not two separate stories.

They are one story told by two teams at the same time.

Let us start with a dangerous misconception

Hemoglobin is late.

In acute postpartum hemorrhage, the hemoglobin value may not immediately show the amount of blood that has been lost.

The local protocol explicitly warns that in PPH, hemoglobin will not reflect rapid blood loss.

So do not wait for a beautiful laboratory confirmation while active bleeding continues.

Use the clinical picture.

Estimated blood loss.

Ongoing active bleeding.

Tachycardia.

Hypotension.

Oliguria.

Altered mental state.

Concealed bleeding.

Coagulopathy.

If the patient is actively hemorrhaging, your resuscitation must be ahead of the lab value.

Now the local definition.

Massive bleeding is defined locally as bleeding in a parturient who required and received two packed cell units and still continues to actively bleed.

That definition is useful because it is operational.

It does not wait for a final total blood loss.

It asks:

She has already needed two packed cell units. Is she still actively bleeding?

If yes, you are in massive active bleeding logic.

Activate the massive hemorrhage protocol.

Notify the blood bank.

Expect packages.

Do not order one unit at a time while the patient is moving faster than the system.

Now let us make the blood products memorable

Packed cells carry oxygen.

Fresh frozen plasma brings clotting factors.

Platelets are the first cellular plugs.

Cryoprecipitate brings concentrated fibrinogen, von Willebrand factor, factor eight, and factor thirteen.

That is the clean version.

Packed cells: oxygen.

Fresh frozen plasma: factors.

Platelets: plugs.

Cryoprecipitate: fibrinogen scaffold.

If you want a construction image, packed cells deliver oxygen workers. Fresh frozen plasma brings tools. Platelets make the first plugs. Cryoprecipitate provides the fibrin scaffold.

But do not let the metaphor become childish.

Return immediately to the clinical reality:

If you give red cells without rebuilding the clot, the circulating volume may look better while the surgical field keeps dissolving.

Packed cells are essential, but packed cells alone can dilute the solution when bleeding is massive.

Gabbe gives useful product facts.

Packed red cells are the only blood product that provides oxygen-carrying capacity. One unit raises hemoglobin by about 1 gram per deciliter in a 70 kilogram patient when the patient is not actively bleeding.

Platelet transfusion should be considered when the platelet count is below 20,000 after vaginal delivery, below 50,000 after cesarean delivery, or when coagulopathy is evident.

Fresh frozen plasma contains clotting factors and also adds volume. Each unit may raise fibrinogen by about 5 to 10 milligrams per deciliter.

Cryoprecipitate is rich in fibrinogen and is especially important when there is hypofibrinogenemia or DIC. It is not a useful volume resuscitation product because each unit is small.

Now the local product strategy.

In massive PPH without increased DIC risk factors, the departmental protocol recommends replacing packed cells and fresh frozen plasma in a one-to-one ratio starting from the third blood unit.

That means that after the initial two packed cell units, ongoing massive bleeding should not continue as red cells alone.

From the third blood unit onward, add coagulation factor replacement.

Add platelets if the platelet count is below 50,000, or after the sixth blood unit.

Add 10 units of cryoprecipitate to the sixth blood unit, after 4 units of fresh frozen plasma.

Monitor fibrinogen, platelets, and coagulation function immediately after the first two packed cells.

Now let us translate the local package logic into speech

Moderate bleeding: two packed cell units.

Massive package one: four packed cells and four fresh frozen plasma units.

Massive package two: two packed cells, two fresh frozen plasma units, and six platelet units.

Massive package three: two packed cells, two fresh frozen plasma units, and 10 cryoprecipitate units.

Then the sequence continues alternating additional packed cells, fresh frozen plasma, platelets, and cryoprecipitate according to the protocol and the blood bank packages.

The exact package table should be available in the clinical protocol during a real event.

For the resident's memory, the principle is more important:

Do not chase massive obstetric bleeding with red cells alone.

Bring clotting factors early.

Bring platelets when indicated.

Bring fibrinogen replacement when needed.

Now DIC

DIC means disseminated intravascular coagulation.

It is a consumptive coagulopathy.

The patient is bleeding because the system that should clot has been activated, consumed, and destabilized.

In obstetrics, some situations tend to produce DIC earlier than ordinary trauma patterns.

The local protocol names amniotic fluid embolism, severe preeclampsia with HELLP syndrome, placental abruption, and chorioamnionitis or sepsis.

Gabbe also includes fetal demise, massive antepartum or postpartum hemorrhage, septic abortion, and acute fatty liver of pregnancy among associated conditions.

The bedside clues are not subtle once you know what to look for.

Diffuse oozing.

Bleeding from puncture sites.

Bleeding from raw surfaces.

Hypotension out of proportion to visible blood loss.

Microangiopathic hemolysis.

Acute lung injury.

Acute renal failure.

A patient who looks much sicker than the visible bleeding explains.

Laboratory confirmation includes low platelets, low fibrinogen, elevated fibrin degradation products, prolonged prothrombin time, prolonged activated partial thromboplastin time, and hemolytic changes when present.

Fibrinogen matters early.

In obstetric hemorrhage, falling fibrinogen is a warning that the hemorrhage is becoming severe and coagulopathic.

Send it early.

Trend it.

Do not wait until the field is all oozing to wonder what the fibrinogen was.

If timely labs are unavailable, Gabbe describes a rough bedside clot test.

Draw 5 milliliters of maternal blood into an empty red-top tube and watch.

If no clot is visible within 6 minutes, or if a clot forms and lyses within 30 minutes, the fibrinogen is usually below 150 milligrams per deciliter.

This is not elegant.

It is not a substitute for labs.

But it is a useful warning in a system delay.

Now the physiologic targets from Gabbe for coagulopathy treatment.

Aim for hematocrit above 21 percent.

Platelets above 50,000 per cubic millimeter.

Fibrinogen above 100 milligrams per deciliter.

Prothrombin time and activated partial thromboplastin time less than 1.5 times control.

Adequate oxygenation.

Normothermia.

And remember the local Stage three instruction: maintain oxygen saturation above 95 percent, maintain body temperature, correct acidosis, check calcium and potassium every 15 minutes during major transfusion, and aggressively treat hypocalcemia and hyperkalemia.

Calcium is not intensive care trivia.

Stored blood products contain citrate, and citrate binds calcium.

Hypocalcemia can worsen hypotension, impair cardiac function, and impair coagulation.

If you replace volume but ignore calcium, you may be resuscitating with one hand and weakening the physiology with the other.

Potassium is also not trivia.

Stored red cells can leak potassium into plasma.

Massive transfusion can cause hyperkalemia.

Hyperkalemia can kill quickly through arrhythmia.

Temperature is not comfort.

Hypothermia impairs coagulation and can cause arrhythmia.

Warm blood products when possible.

Use warming devices.

Cover the patient.

Warm the room if needed.

Acidosis is not just a number.

Acidosis makes clotting and hemodynamics worse.

So the practical massive hemorrhage phrase is:

Blood, clot, calcium, potassium, temperature, acid.

Blood, meaning oxygen-carrying packed cells.

Clot, meaning plasma, platelets, fibrinogen, and correction of coagulopathy.

Calcium, because citrate can drop it.

Potassium, because transfusion can raise it.

Temperature, because cold blood does not clot well.

Acid, because acidosis makes everything harder.

Now recombinant activated factor seven A, also known as NovoSeven.

This is rescue therapy.

It is not routine PPH treatment.

It is not magic dust.

It is hematology-directed rescue in refractory massive bleeding.

The local protocol says it should be used after consultation with the on-call coagulation specialist, at the dose and stage they recommend.

It is considered in uncontrollable massive bleeding after at least 6 blood units, when there is no surgical or angiographic option to stop the bleeding, or bleeding remains refractory.

It requires substrate.

The local text lists minimum conditions: platelets above 30,000, fibrinogen above 50 milligrams per deciliter, and p H above 7.0.

It also recommends preparation with platelets, fresh frozen plasma, and aggressive correction of acidosis when labs are delayed and coagulopathy is hard to control.

Why substrate?

Because factor seven A can amplify thrombin generation, but it cannot build a stable clot out of nothing.

If there are no platelets, no fibrinogen, and severe acidosis, the drug has poor ground to work on.

The local protocol also emphasizes a short hemostatic window.

During the 15 to 30 dry minutes after treatment, reassess and perform surgical hemostasis.

Use the window.

Do not admire it.

The risk is thrombosis.

Reported life-threatening thrombotic events include stroke, myocardial infarction, pulmonary embolism, and deep vein thrombosis.

Avoid it when there is recent thromboembolic disease, evidence of brain death, or no chance of survival for other reasons.

So the NovoSeven script is:

Hematology-directed rescue. Needs platelets, fibrinogen, and p H. Use the short window for hemostasis. Respect thrombosis risk.

Let us now run a clinical sequence.

A woman has ongoing PPH.

She has received two packed cell units and is still actively bleeding.

That meets the local massive bleeding definition.

Activate massive hemorrhage.

Notify the blood bank.

Do not keep ordering single units.

Continue source control.

If atony persists, escalate with tamponade or surgery as appropriate.

If trauma is present, repair it.

If accreta or rupture is present, prepare definitive surgery.

Meanwhile, rebuild the blood.

Packed cells for oxygen.

Fresh frozen plasma for clotting factors.

Platelets for cellular plugs.

Cryoprecipitate for fibrinogen.

Check fibrinogen, platelets, and coagulation studies after the first two packed cells and serially thereafter.

Watch calcium and potassium every 15 minutes during major transfusion.

Warm the patient.

Correct acidosis.

Maintain saturation above 95 percent.

If DIC risk is present, think even earlier about clotting factor and fibrinogen replacement.

If the bleeding remains refractory after major transfusion and source-control options are exhausted or delayed, involve hematology for recombinant activated factor seven A.

Now the closing memory.

Hemoglobin is late.

Red cells carry oxygen, but red cells alone do not rebuild clot.

Fibrinogen matters early.

Platelets matter when low or consumed.

Fresh frozen plasma brings clotting factors.

Cryoprecipitate brings concentrated fibrinogen.

Calcium and potassium can decide whether the heart and clotting system function.

Temperature and acidosis can decide whether hemostasis works.

NovoSeven is rescue, not magic.

And coagulopathy treatment is not after source control.

It happens alongside source control.

In the next episode, we will step backward in time and ask how some hemorrhages should be prevented before they start: cesarean bleeding, placenta previa, placenta accreta spectrum, planned blood bank preparation, and the discipline of treating accreta as a planning diagnosis rather than a surprise at the placental bed.